



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
10-07-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH

PAPER-2

Time : 2:45 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

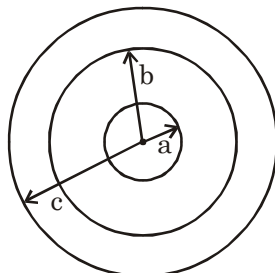
PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 32)

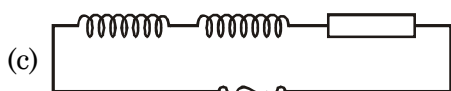
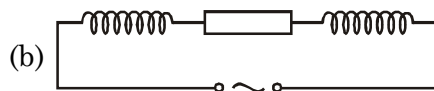
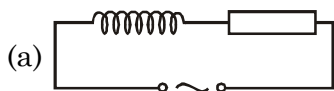
- This section contains **EIGHT** questions.
 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
 - For each question, choose the correct option(s) to answer the question.
 - Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
 - **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.
-
1. The center of mass of a sports car is located at equal distances from the front and rear wheels. If the brake pads are pushed hard so that only the rear wheels only are locked, the stopping distance is L_1 . If the front wheels are locked, the stopping distance is L_2 (At the same initial speed of the car). The other set of wheels roll purely. Neglect the moment of inertia of the wheels.
- (A) $L_1 > L_2$
 (B) $L_2 > L_1$
 (C) When both front and rear wheel are locked, the stopping distance is $(L_1 + L_2)/4$
 (D) When both front and rear wheel are locked, the stopping distance is $(L_1 + L_2)/2$
-

Space for Rough Work

2. Three highly conducting concentric spherical shells (acting as black body) of radii a , b and c are kept in vacuum. The region between the shells is evacuated and they can exchange energy only by radiation. Inside the innermost shell there is a heater of constant power P . In steady state, (σ = Stefan's constant)



- (A) Temperature of outermost shell is $\left(\frac{P}{4\pi\sigma c^2}\right)^{1/4}$
- (B) Temperature of middle shell is $\left(\frac{P}{4\pi\sigma}\right)^{1/4} \left(\frac{1}{c^2} + \frac{1}{b^2}\right)^{1/4}$
- (C) Temperature of innermost shell is $\left(\frac{P}{4\pi\sigma}\right)^{1/4} \left(\frac{1}{a^2} - \frac{1}{b^2}\right)^{1/4}$
- (D) If both outer and middle shell were removed, the temperature of innermost shell would decrease.
3. In circuit (a), the phase difference between the voltage and current through the source is 45° . In circuit (b) and (c) we introduce another identical inductor in series. Since coils are far apart in case (b), we can neglect their mutual inductance but not in case (c). If the phase difference between the voltage and current through the source are ϕ_b and ϕ_c respectively, which of the following is a correct relationship between them?



- (A) $\phi_b > \phi_c$ (B) $\phi_b < \phi_c$ (C) $\phi_b > 45^\circ$ (D) $45^\circ > \phi_c$

Space for Rough Work

4. We have 4 sonometer wires

1 :- length = 4m, mass = 1 gm, T = 100 N

2 :- length = 2m, mass = 2gm, T = 50 N

3 :- length = 8m, mass = 1gm, T = 400 N

4 :- length = 3m, mass = 1.5 gm, T = 600 N

A pulse is generated in each of the string at one of it's end. v is the speed of the wave, t is time taken by the pulse to reach the other end.

(A) $v_1 < v_2 < v_3 = v_4$

(B) $v_2 < v_1 < v_4 < v_3$

(C) $t_1 = t_2 = t_3 > t_4$

(D) $t_4 < t_3 < t_1 < t_2$

5. An equilateral prism has a minimum deviation of 30° .

(A) If the angle of incidence is increased by 1° , angle of emergence would decrease by approx 1°

(B) If the angle of incidence is made $\frac{1}{3}$ rd, the ray would be totally reflected from the second face.

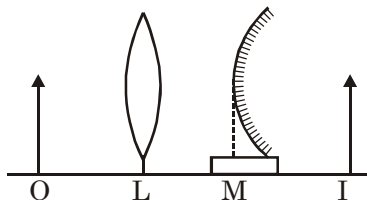
(C) If the angle of incidence is doubled, the ray would be totally reflected from the second face.

(D) $\mu = \sqrt{2}$

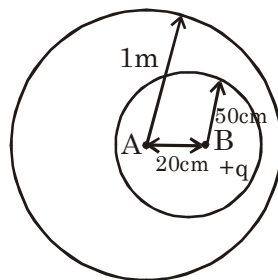
Space for Rough Work

ALLEN

6. An optical bench is used to estimate the focal length of a convex mirror. First an object needle (O), convex lens L and image needle I are placed & parallax is removed. Now convex mirror is introduced between L & I and the mirror is moved so that image as seen from left of O has no parallax with O. Which of the following readings is/are required to estimate the focal length of mirror and random error in it ?



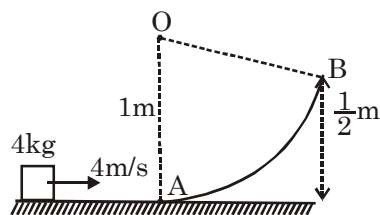
- (A) O (B) L (C) M (D) I
7. A metallic sphere of radius 1m has a spherical cavity of radius $\frac{1}{2}$ m in it as shown. The centers are at a distance of 20 cm from each other. A charge of $q = +5\mu\text{C}$ is kept in the cavity on the line joining the centers at a distance of 40 cm from centre of the original sphere. If net charge on the metallic sphere is $+4\mu\text{C}$, then



- (A) Potential at centre A is 103.5 kV.
 (B) Potential at centre B of cavity is 216 kV.
 (C) The electric field at B is 1125 kN/C.
 (D) The electric field outside the sphere does not change if the position of $+q$ inside cavity changes.

Space for Rough Work

8. A small block of mass 4 kg is projected with a velocity of 4 m/s towards a smooth circular track on smooth ground.



- (A) The normal force just after A is 2.6 times the normal force just before A.
 (B) The speed just before leaving circular track is $\sqrt{6}$ m/s
 (C) The maximum height reached by the block from point A is 0.8 m.
 (D) Angle AOB is 60° .

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

9. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Imagine that we have a bob of mass m attached to a light inextensible string of length L . The other end of the string is tied to the roof. Keeping the string taut, the bob is pulled aside so that the string makes an acute angle θ with the vertical. The bob is then released. Neglect air resistance.

List-I

- (I) Time period of oscillation
 (II) Tension in the string when the string is vertical
 (III) Angular acceleration at the extreme position
 (IV) Kinetic energy of the bob at the mean position

List-II

- (P) Depends on θ if θ is small
 (Q) Depends on θ if θ is large
 (R) Depends on mass m if θ is small
 (S) Depends on L
 (T) is higher if g is higher
 (U) None of these

- (A) I $\rightarrow$ QS; II $\rightarrow$ PQRT; III $\rightarrow$ PQST; IV $\rightarrow$ PQRST
 (B) I $\rightarrow$ QS; II $\rightarrow$ PQRST; III $\rightarrow$ PQRST; IV $\rightarrow$ PQRST
 (C) I $\rightarrow$ PQS; II $\rightarrow$ PQST; III $\rightarrow$ PRST; IV $\rightarrow$ RST
 (D) I $\rightarrow$ QS; II $\rightarrow$ PQST; III $\rightarrow$ PQRT; IV $\rightarrow$ PQRST

Space for Rough Work

10. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Imagine that we have a bob of mass m attached to a light inextensible string of length L . The other end of the string is tied to the roof. Keeping the string taut, the bob is pulled aside so that the string makes an acute angle θ with the vertical. The bob is then released. Neglect air resistance.

List-I

- (I) Acceleration at the mean position
- (II) Acceleration at the extreme position
- (III) Tension in the string at the extreme position
- (IV) Mechanical energy of oscillation

List-II

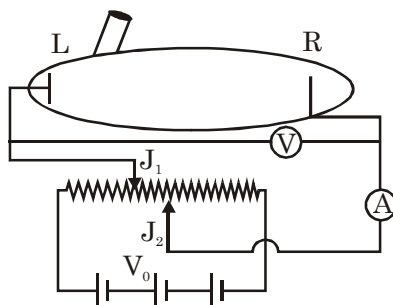
- (P) Depends on θ if θ is small
- (Q) Depends on θ if θ is large
- (R) Depends on mass m if θ is small
- (S) Depends on L
- (T) Is higher if g is higher
- (U) None of these

- (A) I $\rightarrow$ PQST; II $\rightarrow$ PRT; III $\rightarrow$ PQRT; IV $\rightarrow$ PQRST
- (B) I $\rightarrow$ PST; II $\rightarrow$ PQRT; III $\rightarrow$ PRST; IV $\rightarrow$ PQRST
- (C) I $\rightarrow$ PQT; II $\rightarrow$ PQT; III $\rightarrow$ QRT; IV $\rightarrow$ PQRST
- (D) I $\rightarrow$ PT; II $\rightarrow$ PST; III $\rightarrow$ PQRT; IV $\rightarrow$ PQRST

Space for Rough Work

11. Answer the following by appropriately matching the lists based on the information given in the paragraph.

An evacuated tube is set up as shown. Voltmeter and ammeter are ideal. The left plate and right plate are of same metal. Light is incident on the left plate. The work function is 2.5 eV & wavelength of light incident is 414 nm. Assume that if potential difference is accelerating, current is constant. List-I indicates some modification to the set up and list-II indicates the possible effects. Match the appropriate entries.



List-I

- (I) Jockey J_1 is moved to the right of J_2
- (II) Jockey J_1 is moved further left
- (III) Intensity of incident light is increased without changing the wavelength
- (IV) Wavelength of incident light is increased by 10% without changing the intensity.

List-II

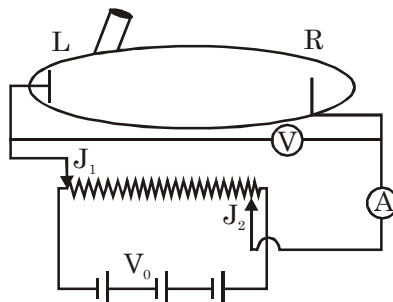
- (P) The reading of ammeter increases.
- (Q) The reading of ammeter may drop to zero.
- (R) The reading of ammeter may decrease.
- (S) The reading of ammeter may remain same.
- (T) The reading of voltmeter may decrease.
- (U) The maximum kinetic energy of incident electrons on plate R may decrease.

- (A) I $\rightarrow$ P,T; II $\rightarrow$ Q,R,S,U; III $\rightarrow$ P; IV $\rightarrow$ P,Q
- (B) I $\rightarrow$ P,S,T; II $\rightarrow$ Q,R,T,U; III $\rightarrow$ P; IV $\rightarrow$ P,Q
- (C) I $\rightarrow$ P,T; II $\rightarrow$ Q,R,U; III $\rightarrow$ S; IV $\rightarrow$ Q,R,S,U
- (D) I $\rightarrow$ P,T; II $\rightarrow$ Q,R,U; III $\rightarrow$ Q; IV $\rightarrow$ P,Q,R

Space for Rough Work

12. Answer the following by appropriately matching the lists based on the information given in the paragraph.

An evacuated tube is set up as shown. Voltmeter and ammeter are ideal. The left plate and right plate are of same metal. Light is incident on the left plate. The work function is 2.5 eV & wavelength of light incident is 414 nm. In this set up voltage $V_0 \gg$ work function and J_1 is far left and J_2 is far right. Electromagnetic radiations are seen to be emitted by plate R. (This situation refers to original case.)



List-I

- (I) V_0 is increased
- (II) The plate R and L are replaced by another metal plate having higher atomic number but same work function.
- (III) Wavelength of incident light is increased without changing the intensity.
- (IV) Wavelength of incident light is decreased without changing the intensity.

List-II

- (P) The radiations emitted will have a minimum wavelength which will be lesser than original case.
- (Q) The radiation emitted may show sharp peaks in intensity which were not seen originally
- (R) Certain sharp peaks in intensity originally seen may be now absent
- (S) The radiations emitted will have a minimum wavelength which will be more than the original case.
- (T) Reading of ammeter may increase.
- (U) Reading of ammeter may decrease.

- (A) I $\rightarrow$ P,Q,T; II $\rightarrow$ Q; III $\rightarrow$ R,S,T; IV $\rightarrow$ P,Q,U
- (B) I $\rightarrow$ P,Q,T; II $\rightarrow$ R; III $\rightarrow$ S,U; IV $\rightarrow$ Q,U
- (C) I $\rightarrow$ P,Q; II $\rightarrow$ R; III $\rightarrow$ R,S,U; IV $\rightarrow$ P,U
- (D) I $\rightarrow$ P,Q; II $\rightarrow$ Q,R; III $\rightarrow$ R,S,T,U; IV $\rightarrow$ P,Q,U

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

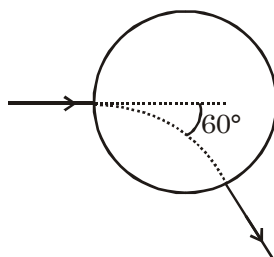
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	●	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

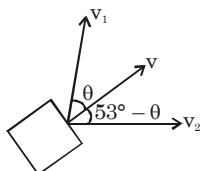
-									
●	●	●	0	0	●	●	●	●	●
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	●	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Current is flowing in an ideal solenoid. When an electron is projected towards its axis, it does not cross the axis but gets deflected by 60° in course of its passage. If solenoid has 200 turns/m, find current I (in A) in solenoid. Assume current to be positive if it is clockwise as seen from above and negative if it is anticlockwise as seen from above. Radius of solenoid is 1m. Speed of projection is $\frac{\sqrt{3}}{2}$ m/s. Here $m = 9 \times 10^{-31}$ kg. Fill $I \times 10^8$ in OMR.

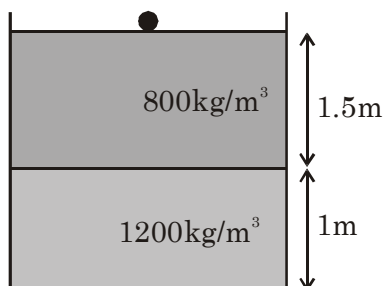


- A box is being pulled by two tractors. One of the tractors has a velocity of $v_1 = 5$ m/s and other has velocity of $v_2 = 3$ m/s. The ropes are parallel to velocity vectors. Find the velocity of the box (in m/s). Box is not rotating.



Space for Rough Work

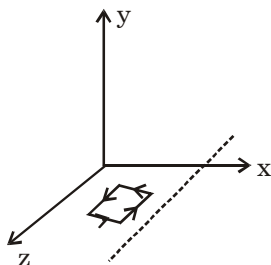
3. A small ball is dropped at the surface of the vessel containing two immiscible liquids as shown. The velocity of ball becomes zero just as it reaches the bottom. What is the density (in kg/m^3) of the ball ? Neglect viscosity.



4. The de-broglie wavelength of a particle as seen from an inertial frame is $25\text{ }\mu\text{m}$ and as seen from another inertial frame moving in opposite direction to the first (but with same speed) is $20\text{ }\mu\text{m}$. The mass of particle is $6.63 \times 10^{-30}\text{ kg}$. What is its de-broglie wavelength (in μm) as seen from ground frame ? The velocity of all three (particle and both frames) is along the same line. (Speed of particle is greater than either of frames)

Space for Rough Work

5. On a smooth horizontal ground, we kept a square loop of mass 2 kg having sides 20×20 cm. Sides are parallel to x & z axis. Beyond $x = +1$ m, uniform magnetic field of $2\hat{i} + 4\hat{j} + 6\hat{k}$ (T) is present. The loop enters the magnetic field with negligible velocity. If the current in the loop is held constant at 1A, what is the velocity (in m/s) of the loop when it has completely entered the region of the magnetic field ?



6. The horizontal plates of a capacitor are charged to a voltage of 20 V. A particle carrying a charge is in equilibrium under the influence of gravitational and electrostatic forces. Capacitor is now connected to a resistance $R = 4\text{k}\Omega$. If capacitance is $500 \mu\text{F}$, then find the velocity (in m/s) of the particle at $t = 2$ sec.

Space for Rough Work

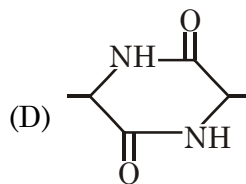
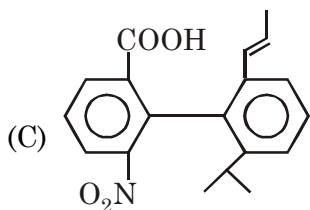
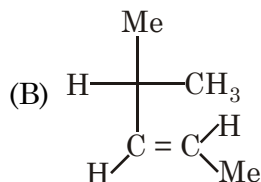
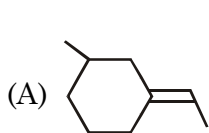
PART-2 : CHEMISTRY**SECTION-I(i) : (Maximum Marks: 32)**

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

-
1. Choose **INCORRECT** order
 (A) Thermal stability : $\text{BeO}_2 < \text{MgO}_2 < \text{CaO}_2 < \text{SrO}_2 < \text{BaO}_2$
 (B) Solubility in water : $\text{LiF} > \text{NaF} > \text{KF} > \text{RbF} > \text{CsF}$
 (C) Ionic character : $\text{BeF}_2 < \text{MgF}_2 < \text{CaF}_2 < \text{SrF}_2 < \text{BaF}_2$
 (D) Basic nature : $\text{BeO} < \text{MgO} < \text{CaO} < \text{SrO} < \text{BaO}$
 2. Which of the following ore is paramagnetic in nature?
 (A) Bauxite (B) Chromite ore (C) Cryolite (D) Zinc blende
 3. In which of the following paramagnetic complex have pairing energy (P) is lower than splitting energy (Δ).
 (A) $[\text{NiF}_6]^{2-}$ (B) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ (C) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ (D) $[\text{RuF}_6]^{3-}$
-

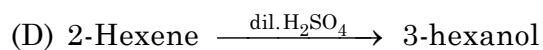
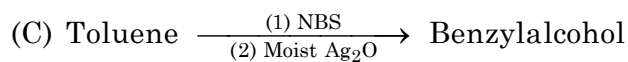
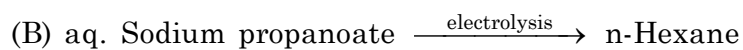
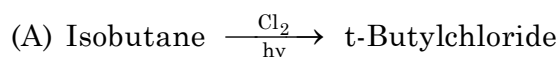
Space for Rough Work

4. One mole water is completely converted into water vapour at 27°C and 0.04 atm .
 $\text{H}_2\text{O}(l, 27^{\circ}\text{C}, 0.04\text{ atm}) \rightarrow \text{H}_2\text{O}(g, 27^{\circ}\text{C}, 0.04\text{ atm})$.
 The vapour pressure of $\text{H}_2\text{O}(l)$ at 27°C is 30.4 torr . The correct value (s) for the given parameters for this change is /are -
 (A) $\Delta H_{\text{sys}} = +ve$ (B) $\Delta S_{\text{sys}} = +ve$ (C) $\Delta G_{\text{sys}} = 0$ (D) $\Delta S_{\text{univ.}} = +ve$
5. A cubic crystal is FCC for 'A' atoms and 'B' atoms occupy the body centres and all the edge-centres. The correct information(s) about this solid is/are : (Given $r_B = (\sqrt{2} - 1)r_A$)
 (A) The simplest formula of solid is AB.
 (B) There are 4 'AB' formula units per unit cell.
 (C) Two 'B' atoms can touch each other.
 (D) Two 'A' atoms can touch each other.
6. Which at the following is/are capable of showing geometrical as well as optical isomerism ?

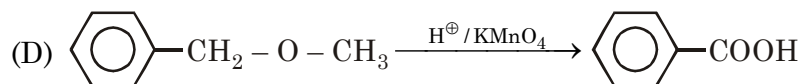
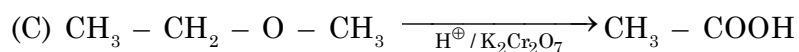
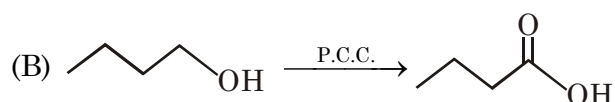
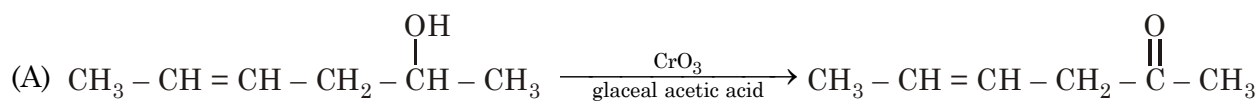


Space for Rough Work

7. Select reaction(s) with correct major product ?



8. Select correct reaction(s) showing major product



Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

9. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Cyclic carbohydrates which has anomeric -OH gives Tollen's test and also show mutarotation.

List-I		List-II	
(I)	Maltose	(P)	It can show mutarotation
(II)	Sucrose	(Q)	It gives Tollen's test
(III)	Cellulose	(R)	α -D-glucose is one of the monomer
(IV)	Lactose	(S)	It has 1,2-Glycoside linkage
		(T)	β -D-glucose is one of the monomer
		(U)	It is not a reducing saccharide

Which of the following options has all correct combination considering list-I and list-II

- (A) I, P,Q,R,S (B) II, S,T,U (C) III, S,T,U (D) IV, P,Q,T

Space for Rough Work

ALLEN

10. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Cyclic carbohydrates which has anomeric –OH gives Tollen's test and also show mutarotation.

List-I	List-II
(I) Maltose	(P) It can show mutarotation
(II) Sucrose	(Q) It gives Tollen's test
(III) Cellulose	(R) α -D-glucose is one of the monomer
(IV) Lactose	(S) It has 1,2-Glycoside linkage
	(T) β -D-glucose is one of the monomer
	(U) It is not a reducing saccharide

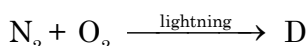
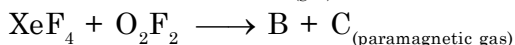
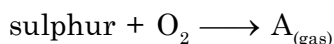
Which of the following option has incorrect combination considering

- (A) I, P,Q,R (B) II, P,S,T,U (C) III, T,U (D) IV, P,Q,T

11. Answer the following by appropriately matching the lists based on the information given in the paragraph.

The properties of p-block elements like that of others are greatly influenced by atomic sizes, ionisation enthalpy, electron gain enthalpy and electronegativity. The absence of d-orbitals in second period and presence of d or d and f orbitals in heavier elements (starting from third period onwards) have significant effects on the properties of elements. In addition, the presence of all the three types of elements; metals, metalloids and non-metals bring diversification in chemistry of these elements.

Consider the following reactions



List-I	List-II
(I) (A) + $\text{Cl}_2 \xrightarrow{\text{charcoal}}$ product(s)	(P) any product is paramagnetic
(II) (B) + $\xrightarrow{\text{complete hydrolysis}}$ product(s)	(Q) All products are planar
(III) D + C $\longrightarrow$ product(s)	(R) All products are polar
(IV) A + C $\xrightarrow[\text{low temp. \& high pressure}]{\text{catalyst}}$ product(s)	(S) In any product p_π - p_π bond present
	(T) Any product is explosive
	(U) Any product is coloured

Which of the following option has the **INCORRECT** combination considering List-I and List-II ?

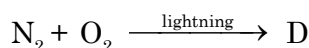
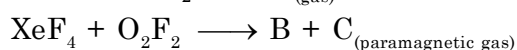
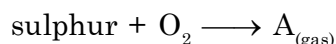
- (A) I, Q (B) II, R (C) III, U (D) IV, Q

Space for Rough Work

12. Answer the following by appropriately matching the lists based on the information given in the paragraph.

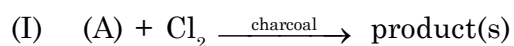
The properties of p-block elements like that of others are greatly influenced by atomic sizes, ionisation enthalpy, electron gain enthalpy and electronegativity. The absence of d-orbitals in second period and presence of d or d and f orbitals in heavier elements (starting from third period onwards) have significant effects on the properties of elements. In addition, the presence of all the three types of elements; metals, metalloids and non-metals bring diversification in chemistry of these elements.

Consider the following reactions

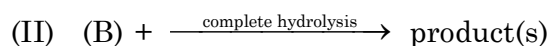


List-I

List-II



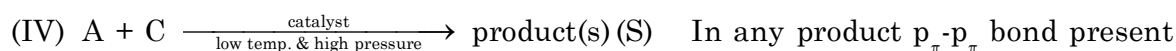
(P) any product is paramagnetic



(Q) All products are planar



(R) All products are polar



(T) Any product is explosive

(U) Any product is coloured

Which of the following option has the correct combination considering List-I and List-II ?

(A) I, P

(B) II, Q

(C) III, S

(D) IV, U

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	●	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

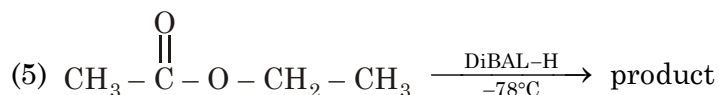
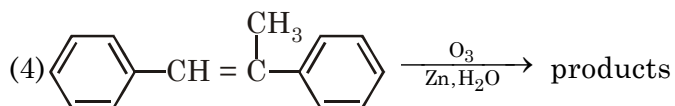
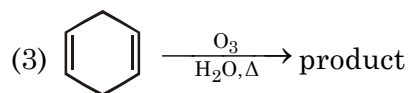
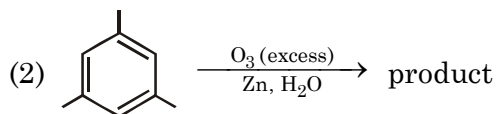
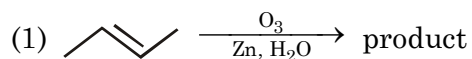
-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Five moles of a vanderwaal gas occupy 5.0 litre at 177°C and 35.5 atm. If the value of 'b' is 0.1 L mol⁻¹, the value of 'a' (in atm L²mol⁻²) is (R = 0.082L-atm/K-mol)
- E_{cell} for the following Galvanic cell at 298K is (x × 10⁻²)V.
 Pt(s) | H₂(g, 2.0 bar) | HA (0.2M, K_a = 2 × 10⁻⁵) || HB(0.1M, K_a = 4 × 10⁻⁶) | H₂(g, 0.4 bar) | Pt(s).
 The value of 'x' is.
 (Given : $\frac{2.303R \times 298}{F} = 0.06V, \log 2 = 0.3$)
- The specific activity of carbon in wooden samples 'A' and 'B' are 5000 counts per min per gm and 4000 counts per min per gm, respectively. How much older (in years) the sample 'B' is than the sample 'A' ? (t_{1/2} of C¹⁴ = 5775 yrs, ln 5 = 1.6, ln10 = 2.3)

Space for Rough Work

4. In H-like atoms, the number of orbitals having 2 radial as well as 2 angular nodes is.
5. In how many of following reaction(s), at least one product gives +ve test with Fehling solution and also gives Iodoform test



6. Total number of species which give O_2 when react with H_2O_2
 MnO_4^- , PbS , NH_2OH , $[\text{Fe}(\text{CN})_6]^{3-}$ (basic medium), SO_3^{2-} , HOCl

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks: 32)**

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

-
1. Consider the equation, $\operatorname{cosec}\left(\frac{\pi}{4} + x\right) + \operatorname{cosec}\left(\frac{\pi}{4} - x\right) = 2\sqrt{2}$
(A) Number of solution of the equation in $x \in [0, 4\pi]$ is 7
(B) Number of solution of the equation in $x \in [0, 4\pi]$ is 6
(C) Number of solution of the equation in $x \in (0, 2\pi)$ is 2
(D) Number of solution of the equation in $x \in (0, 2\pi)$ is 3
 2. Triangle ABC is right angle at A. The point P and Q are on side BC such that $BP = PQ = QC$. If $\ell(AP) = 3$ and $\ell(AQ) = 4$, then $\ell(BC)$ equals
(A) $\sqrt{27}$ (B) $\sqrt{36}$ (C) $\sqrt{45}$ (D) $\sqrt{54}$
 3. Tangent to parabola $y^2 = 8x$ are drawn from the point $P(6, 8)$. The point of contact of these tangents are Q and R. (S is the focus of the parabola). Which of the following is/are true
(A) Mid point of QR is (10, 8)
(B) Area of ΔPQR is 16 sq. units
(C) Equation of circle circumscribing the ΔPQR is $x^2 + y^2 - 36x + 16y - 18 = 0$
(D) Equation of circle circumscribing the ΔPQR is $x^2 + y^2 - 36x + 16y - 12 = 0$

Space for Rough Work

4. Let $A(z_1)$, $B(z_2)$ and $C(z_3)$ lie on the circle $|z - i| = 1$ and satisfy the equation $3z_1 + i = 2z_2 + 2z_3$. If D is the centre of the circle $|z - i| = 1$, then which of the following is/are true
- (A) Area of quadrilateral $ABCD$ is $\frac{\sqrt{7}}{4}$ (B) Area of quadrilateral $ABCD$ is $\frac{\sqrt{11}}{4}$
- (C) $\ell(BC) = \frac{\sqrt{7}}{2}$ (D) $\ell(BC) = \frac{\sqrt{11}}{2}$
5. Let $P_n = P_{n-1} + P_{n-2} \forall n \geq 2$ and $n \in \mathbb{N}$ where $P_0 = 0$ and $P_1 = 1$, then which of the following is/are true
- (A) $\sum_{k=1}^{\infty} \frac{P_k}{3^k} = \frac{3}{5}$ (B) $\sum_{k=1}^{\infty} \frac{P_k}{3^k} = \frac{2}{5}$ (C) $P_9 = 34$ (D) $P_9 = 55$
6. The probability that a married man watches a certain TV show is 0.4 and probability that a married women watches the show is 0.5. The probability that man watches the show, given that his wife does, is 0.7. Then which of the following is/are true
- (A) The probability that married couple watches the show is 0.45
- (B) The probability that a wife watches the show given that her husband does is $\frac{7}{8}$
- (C) The probability that married couple watches the show is 0.35
- (D) The probability that atleast one person of the married couple will watch the show is 0.55
7. Let $a, b > 0$ and $\Delta = \begin{vmatrix} -x & a & b \\ b & -x & a \\ a & b & -x \end{vmatrix}$, then which of the following is/are true
- (A) $(a + b - x)$ is a factor of Δ (B) $x^2 + (a + b)x + (a^2 + b^2 + ab)$, is a factor of Δ
- (C) $\Delta = 0$ has three real roots if $a = b$ (D) $\Delta = 0$ has one real root if $a = b$
8. If A and B are square matrices of order '3' such that $A \operatorname{adj}(2B) = 16I_3$ and $\det(B) = 2$ then which of the following is/are true (Where $\operatorname{adj}(X)$ denotes adjoint of matrix (X) , X^{-1} denotes inverse of matrix (X)) and $\det(X)$ denotes determinant value of matrix (X) .
- (A) $(A^{-1}(\operatorname{adj}B))^{-1} = 4A^2$ (B) $(A^{-1}(\operatorname{adj}B))^{-1} = B^2$
- (C) $B(\operatorname{adj} A) = 8I_3$ (D) $B(\operatorname{adj} A) = 4I_3$

Space for Rough Work

ALLEN**SECTION-I(ii) : (Maximum Marks : 12)**

- This section contains **TWO** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

9. Answer the following by appropriately matching the lists based on the information given in the paragraph

Let $I_n = \lim_{x \rightarrow \infty} \int_{e^{-x}}^1 \left(\log_e \frac{1}{t} \right)^n dt$, where $n \in \mathbb{N}$.

Let $J = \int_0^{1/2} \left(\frac{1}{4} - x^2 \right)^4 dx$ and $K = \int_0^{1/2} x^4 (1-x)^4 dx$.

There are some expressions given in List-I whose values are given in List-II given below

List-I	List-II
(I) $\frac{J}{K}$	(P) 4
(II) $\frac{I_5}{I_4}$	(Q) 1
(III) $\frac{I_{100}}{25I_{99}}$	(R) 0
(IV) 3780 J	(S) 2
	(T) 5
	(U) 3

Which of the following is the only CORRECT combination ?

- (A) (II), (T) (B) (I), (S) (C) (III), (U) (D) (IV), (P)

Space for Rough Work

10. Answer the following by appropriately matching the lists based on the information given in the paragraph

Let $I_n = \lim_{x \rightarrow \infty} \int_{e^{-x}}^1 \left(\log_e \frac{1}{t} \right)^n dt$, where $n \in \mathbb{N}$.

Let $J = \int_0^{1/2} \left(\frac{1}{4} - x^2 \right)^4 dx$ and $K = \int_0^{1/2} x^4 (1-x)^4 dx$.

There are some expression given in the list-I whose values are given in List-II given below

List-I	List-II
(I) $\frac{J}{K}$	(P) 4
(II) $\frac{I_5}{I_4}$	(Q) 1
(III) $\frac{I_{100}}{25I_{99}}$	(R) 0
(IV) 3780 J	(S) 2
	(T) 5
	(U) 3

Which of the following is the only CORRECT combination ?

- (A) (II), (P) (B) (IV), (U) (C) (III), (S) (D) (I), (S)

Space for Rough Work

11. Answer the following by appropriately matching the lists based on the information given in the paragraph

Consider the planes

$$P_1 : x + y - z = 5$$

$$P_2 : 2x - y + \lambda z = 3 \quad (\text{where } \lambda \in \mathbb{R})$$

$$P_3 : x - 3y - 4 = 0$$

$$P_4 : 4y - z + 5 = 0$$

Plane P_1 and P_2 intersect to form line L_1 . Plane P_3 and P_4 intersect to form line L_2 .

There are some expression given in the List-I match them with the List-II given below

List-I

List-II

- | | |
|--|--------------------|
| (I) Value of λ for which L_1 and L_2 are coplaner | (P) $-\frac{5}{4}$ |
| (II) Image of the line L_2 in the plane P_1 is passing through | (Q) $(0, -33, 0)$ |
| (III) Image of plane P_1 in the plane P_3 is passing through | (R) $-\frac{6}{5}$ |
| (IV) Equation of the line L_2 is passing through | (S) $(1, -1, 1)$ |
| | (T) $(-1, 1, -11)$ |
| | (U) $(11, 5, -5)$ |

Which of the following is the only CORRECT combination ?

- (A) (I), (P) (B) (II), (S) (C) (III), (U) (D) (IV), (T)

Space for Rough Work

12. Answer the following by appropriately matching the lists based on the information given in the paragraph

Consider the planes

$$P_1 : x + y - z = 5$$

$$P_2 : 2x - y + \lambda z = 3 \quad (\text{where } \lambda \in \mathbb{R})$$

$$P_3 : x - 3y - 4 = 0$$

$$P_4 : 4y - z + 5 = 0$$

Plane P_1 and P_2 intersect to form line L_1 . Plane P_3 and P_4 intersect to form line L_2 .

There are some expression given in the List-I match them with the List-II given below

List-I

List-II

- | | |
|--|--------------------|
| (I) Value of λ for which L_1 and L_2 are coplaner | (P) $-\frac{5}{4}$ |
| (II) Image of the line L_2 in the plane P_1 is passing through | (Q) $(0, -33, 0)$ |
| (III) Image of plane P_1 in the plane P_3 is passing through | (R) $-\frac{6}{5}$ |
| (IV) Equation of the line L_2 is passing through | (S) $(1, -1, 1)$ |
| | (T) $(-1, 1, -11)$ |
| | (U) $(11, 5, -5)$ |

Which of the following is the only INCORRECT combination ?

- (A) (I), (P) (B) (II), (S) (C) (III), (Q) (D) (IV), (S)

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Let $f(x)$ be differentiable function and $|f'(x)| \leq 2$ for all real x . If $f(1) = 2$ and $f(4) = 8$, then the value of $f(2)$ equals
- If the range of values of 'a' for which $f(x) = \log_a(4ax - x^2)$ is strictly increasing $\forall x \in \left[\frac{3}{2}, 2\right]$ is $(p, q] \cup (r, \infty)$ then the value of $(2p + 4q + r)$ equals
- If $\int \frac{(\sec x - \tan x) dx}{\sqrt{\sin^2 x - \sin x}} = K \log_e |f(x) + \sqrt{2} \sqrt{\tan x (\tan x - \sec x)}| + C$, where C is the constant of integration and K is the fixed constant, the value of $\left(K^2 + \left[\left(f\left(\frac{4\pi}{3}\right) \right)^2 \right] \right)$ equals
(Where $[.]$ denotes greatest integer function)

Space for Rough Work

4. Let $y = f(x)$ be a curve which satisfy the differential equation $e^x dy + (x - 1)dx = 0$ and $f(0) = 0$. If the area enclosed by the curve, x-axis, y-axis and a line $x = c$, where $(c, f(c))$ is the point of inflection on the curve is $\left(1 - \frac{p}{e^q}\right)$ ($p, q \in \mathbb{N}$), then the value of $(p + q)$ equals
5. Circle which touches both the axes and the line $4x + 3y = 12$ and lies in the first quadrant. Sum of all the possible radius of such circle is
6. Let $f(x) = \begin{cases} |x|^{p-2} \sin \frac{1}{x} + x |\tan x|^{q-3} & x \neq 0 \\ 0 & x = 0 \end{cases}$, is differentiable at $x = 0$, then minimum value of $[p + q]$ equals (where $[.]$ represents greatest integer function)

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	8	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-1 In all other cases	32
I(ii)	Matching Lists Type (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	12
II	Numerical Value Type (Up to second decimal place)	6	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	18

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator